

SECTION 9

MC68010 INSTRUCTION EXECUTION TIMES

This section contains listings of the instruction execution times in terms of external clock (CLK) periods for the MC68010. In this data, it is assumed that both memory read and write cycles consist of four clock periods. A longer memory cycle causes the generation of wait states that must be added to the total instruction times.

The number of bus read and write cycles for each instruction is also included with the timing data. This data is shown as

$$n(r/w)$$

where:

- n is the total number of clock periods
- r is the number of read cycles
- w is the number of write cycles

For example, a timing number shown as 18(3/1) means that 18 clock cycles are required to execute the instruction. Of the 18 clock periods, 12 are used for the three read cycles (four periods per cycle). Four additional clock periods are used for the single write cycle, for a total of 16 clock periods. The bus is idle for two clock periods during which the processor completes the internal operations required for the instructions.

NOTE

The total number of clock periods (n) includes instruction fetch and all applicable operand fetches and stores.

9.1 OPERAND EFFECTIVE ADDRESS CALCULATION TIMES

Table 9-1 lists the numbers of clock periods required to compute the effective addresses for instructions. The totals include fetching any extension words, computing the address, and fetching the memory operand. The total number of clock periods, the number of read cycles, and the number of write cycles (zero for all effective address calculations) are shown in the previously described format.

Table 9-1. Effective Address Calculation Times

Addressing Mode		Byte, Word		Long	
		Fetch	No Fetch	Fetch	No Fetch
Register					
Dn	Data Register Direct	0(0/0)	—	0(0/0)	—
An	Address Register Direct	0(0/0)	—	0(0/0)	—
Memory					
(An)	Address Register Indirect	4(1/0)	2(0/0)	8(2/0)	2(0/0)
(An)+	Address Register Indirect with Postincrement	4(1/0)	4(0/0)	8(2/0)	4(0/0)
-(An)	Address Register Indirect with Predecrement	6(1/0)	4(0/0)	10(2/0)	4(0/0)
(d ₁₆ , An)	Address Register Indirect with Displacement	8(2/0)	4(0/0)	12(3/0)	4(1/0)
(d ₈ , An, Xn)*	Address Register Indirect with Index	10(2/0)	8(1/0)	14(3/0)	8(1/0)
(xxx).W	Absolute Short	8(2/0)	4(1/0)	12(3/0)	4(1/0)
(xxx).L	Absolute Long	12(3/0)	8(2/0)	16(4/0)	8(2/0)
(d ₁₆ , PC)	Program Counter Indirect with Displacement	8(2/0)	—	12(3/0)	—
(d ₈ , PC, Xn)*	Program Counter Indirect with Index	10(2/0)	—	14(3/0)	—
#<data>	Immediate	4(1/0)	—	8(2/0)	—

*The size of the index register (Xn) does not affect execution time.

9.2 MOVE INSTRUCTION EXECUTION TIMES

Tables 9-2, 9-3, 9-4, and 9-5 list the numbers of clock periods for the move instructions. The totals include instruction fetch, operand reads, and operand writes. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format.

Table 9-2. Move Byte and Word Instruction Execution Times

Source	Destination								
	Dn	An	(An)	(An)+	-(An)	(d16, An)	(dg, An, Xn)*	(xxx).W	(xxx).L
Dn	4(1/0)	4(1/0)	8(1/1)	8(1/1)	8(1/1)	12(2/1)	14(2/1)	12(2/1)	16(3/1)
An	4(1/0)	4(1/0)	8(1/1)	8(1/1)	8(1/1)	12(2/1)	14(2/1)	12(2/1)	16(3/1)
(An)	8(2/0)	8(2/0)	12(2/1)	12(2/1)	12(2/1)	16(3/1)	18(3/1)	16(3/1)	20(4/1)
(An)+	8(2/0)	8(2/0)	12(2/1)	12(2/1)	12(2/1)	16(3/1)	18(3/1)	16(3/1)	20(4/1)
-(An)	10(2/0)	10(2/0)	14(2/1)	14(2/1)	14(2/1)	18(3/1)	20(3/1)	18(3/1)	22(4/1)
(d16, An)	12(3/0)	12(3/0)	16(3/1)	16(3/1)	16(3/1)	20(4/1)	22(4/1)	20(4/1)	24(5/1)
(dg, An, Xn)*	14(3/0)	14(3/0)	18(3/1)	18(3/1)	18(3/1)	22(4/1)	24(4/1)	22(4/1)	26(5/1)
(xxx).W	12(3/0)	12(3/0)	16(3/1)	16(3/1)	16(3/1)	20(4/1)	22(4/1)	20(4/1)	24(5/1)
(xxx).L	16(4/0)	16(4/0)	20(4/1)	20(4/1)	20(4/1)	24(5/1)	26(5/1)	24(5/1)	28(6/1)
(d16, PC)	12(3/0)	12(3/0)	16(3/1)	16(3/1)	16(3/1)	20(4/1)	22(4/1)	20(4/1)	24(5/1)
(dg, PC, Xn)*	14(3/0)	14(3/0)	18(3/1)	18(3/1)	18(3/1)	22(4/1)	24(4/1)	22(4/1)	26(5/1)
#<data>	8(2/0)	8(2/0)	12(2/1)	12(2/1)	12(2/1)	16(3/1)	18(3/1)	16(3/1)	20(4/1)

*The size of the index register (Xn) does not affect execution time.

Table 9-3. Move Byte and Word Instruction Loop Mode Execution Times

Source	Loop Continued			Loop Terminated					
	Valid Count, cc False			Valid count, cc True			Expired Count		
	Destination								
	(An)	(An)+	−(An)	(An)	(An)+	−(An)	(An)	(An)+	−(An)
Dn	10(0/1)	10(0/1)	—	18(2/1)	18(2/1)	—	16(2/1)	16(2/1)	—
An*	10(0/1)	10(0/1)	—	18(2/1)	18(2/1)	—	16(2/1)	16(2/1)	—
(An)	14(1/1)	14(1/1)	16(1/1)	20(3/1)	20(3/1)	22(3/1)	18(3/1)	18(3/1)	20(3/1)
(An)+	14(1/1)	14(1/1)	16(1/1)	20(3/1)	20(3/1)	22(3/1)	18(3/1)	18(3/1)	20(3/1)
−(An)	16(1/1)	16(1/1)	18(1/1)	22(3/1)	22(3/1)	24(3/1)	20(3/1)	20(3/1)	22(3/1)

*Word only.

Table 9-4. Move Long Instruction Execution Times

Source	Destination								
	Dn	An	(An)	(An)+	-(An)	(d16, An)	(dg, An, Xn)*	(xxx).W	(xxx).L
Dn	4(1/0)	4(1/0)	12(1/2)	12(1/2)	14(1/2)	16(2/2)	18(2/2)	16(2/2)	20(3/2)
An	4(1/0)	4(1/0)	12(1/2)	12(1/2)	14(1/2)	16(2/2)	18(2/2)	16(2/2)	20(3/2)
(An)	12(3/0)	12(3/0)	20(3/2)	20(3/2)	20(3/2)	24(4/2)	26(4/2)	24(4/2)	28(5/2)
(An)+	12(3/0)	12(3/0)	20(3/2)	20(3/2)	20(3/2)	24(4/2)	26(4/2)	24(4/2)	28(5/2)
-(An)	14(3/0)	14(3/0)	22(3/2)	22(3/2)	22(3/2)	26(4/2)	28(4/2)	26(4/2)	30(5/2)
(d16, An)	16(4/0)	16(4/0)	24(4/2)	24(4/2)	24(4/2)	28(5/2)	30(5/2)	28(5/2)	32(6/2)
(dg, An, Xn)*	18(4/0)	18(4/0)	26(4/2)	26(4/2)	26(4/2)	30(5/2)	32(5/2)	30(5/2)	34(6/2)
(xxx).W	16(4/0)	16(4/0)	24(4/2)	24(4/2)	24(4/2)	28(5/2)	30(5/2)	28(5/2)	32(6/2)
(xxx).L	20(5/0)	20(5/0)	28(5/2)	28(5/2)	28(5/2)	32(6/2)	34(6/2)	32(6/2)	36(7/2)
(d16, PC)	16(4/0)	16(4/0)	24(4/2)	24(4/2)	24(4/2)	28(5/2)	30(5/2)	28(5/2)	32(5/2)
(dg, PC, Xn)*	18(4/0)	18(4/0)	26(4/2)	26(4/2)	26(4/2)	30(5/2)	32(5/2)	30(5/2)	34(6/2)
#<data>	12(3/0)	12(3/0)	20(3/2)	20(3/2)	20(3/2)	24(4/2)	26(4/2)	24(4/2)	28(5/2)

*The size of the index register (Xn) does not affect execution time.

Table 9-5. Move Long Instruction Loop Mode Execution Times

Source	Loop Continued			Loop Terminated					
	Valid Count, cc False			Valid count, cc True			Expired Count		
	Destination								
	(An)	(An)+	−(An)	(An)	(An)+	−(An)	(An)	(An)+	−(An)
Dn	14(0/2)	14(0/2)	—	20(2/2)	20(2/2)	—	18(2/2)	18(2/2)	—
An	14(0/2)	14(0/2)	—	20(2/2)	20(2/2)	—	18(2/2)	18(2/2)	—
(An)	22(2/2)	22(2/2)	24(2/2)	28(4/2)	28(4/2)	30(4/2)	24(4/2)	24(4/2)	26(4/2)
(An)+	22(2/2)	22(2/2)	24(2/2)	28(4/2)	28(4/2)	30(4/2)	24(4/2)	24(4/2)	26(4/2)
−(An)	24(2/2)	24(2/2)	26(2/2)	30(4/2)	30(4/2)	32(4/2)	26(4/2)	26(4/2)	28(4/2)

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9.3 STANDARD INSTRUCTION EXECUTION TIMES

The numbers of clock periods shown in tables 9-6 and 9-7 indicate the times required to perform the operations, store the results, and read the next instruction. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

In Tables 9-6 and 9-7, the following notation applies:

- An — Address register operand
- Sn — Data register operand
- ea — An operand specified by an effective address
- M — Memory effective address operand

Table 9-6. Standard Instruction Execution Times

Instruction	Size	op<ea>, An***	op<ea>, Dn	op Dn, <M>
ADD/ADDA	Byte, Word	8(1/0)+	4(1/0)+	8(1/1)+
	Long	6(1/0)+	6(1/0)+	12(1/2)+
AND	Byte, Word	—	4(1/0)+	8(1/1)+
	Long	—	6(1/0)+	12(1/2)+
CMP/CMPL	Byte, Word	6(1/0)+	4(1/0)+	—
	Long	6(1/0)+	6(1/0)+	—
DIVS	—	—	122(1/0)+	—
DIVU	—	—	108(1/0)+	—
EOR	Byte, Word	—	4(1/0)**	8(1/1)+
	Long	—	6(1/0)**	12(1/2)+
MULS/MULU	—	—	42(1/0)+*	—
	—	—	40(1/0)*	—
OR	Byte, Word	—	4(1/0)+	8(1/1)+
	Long	—	6(1/0)+	12(1/2)+
SUB/SUBA	Byte, Word	8(1/0)+	4(1/0)+	8(1/1)+
	Long	6(1/0)+	6(1/0)+	12(1/2)+

+ Add effective address calculation time.

* Indicates maximum value.

** Only available address mode is data register direct.

*** Word or long word only.

Table 9-7 Standard Instruction Loop Mode Execution Times

Instruction	Size	Loop Continued			Loop Terminated					
		Valid Count cc False			Valid Count cc True			Expired Count		
		op<ea>, An*	op<ea>, Dn	op Dn, <ea>	op<ea>, An*	op<ea>, Dn	op Dn, <ea>	op<ea>, An*	op<ea>, Dn	op Dn, <ea>
ADD	Byte, Word	18(1/0)	16(1/0)	16(1/1)	24(3/0)	22(3/0)	22(3/1)	22(3/0)	20(3/0)	20(3/1)
	Long	22(2/0)	22(2/0)	24(2/2)	28(4/0)	28(4/0)	30(4/2)	26(4/0)	26(4/0)	28(4/2)
AND	Byte, Word	—	16(1/0)	16(1/1)	—	22(3/0)	22(3/1)	—	20(3/0)	20(3/1)
	Long	—	22(2/0)	24(2/2)	—	28(4/0)	30(4/2)	—	26(4/0)	28(4/2)
CMP	Byte, Word	12(1/0)	12(1/0)	—	18(3/0)	18(3/0)	—	16(3/0)	16(4/0)	—
	Long	18(2/0)	18(2/0)	—	24(4/0)	24(4/0)	—	20(4/0)	20(4/0)	—
EOR	Byte, Word	—	—	16(1/0)	—	—	22(3/1)	—	—	20(3/1)
	Long	—	—	24(2/2)	—	—	30(4/2)	—	—	28(4/2)
OR	Byte, Word	—	16(1/0)	16(1/0)	—	22(3/0)	22(3/1)	—	20(3/0)	20(3/1)
	Long	—	22(2/0)	24(2/2)	—	28(4/0)	30(4/2)	—	26(4/0)	28(4/2)
SUB	Byte, Word	18(1/0)	16(1/0)	16(1/1)	24(3/0)	22(3/0)	22(3/1)	22(3/0)	20(3/0)	20(3/1)
	Long	22(2/0)	20(2/0)	24(2/2)	28(4/0)	26(4/0)	30(4/2)	26(4/0)	24(4/0)	28(4/2)

*Word or long word only.

<ea> may be (An), (An)+, or -(An) only. Add two clock periods to the table value if <ea> is -(An).

9.4 IMMEDIATE INSTRUCTION EXECUTION TIMES

The numbers of clock periods shown in Table 9-8 include the times to fetch immediate operands, perform the operations, store the results, and read the next operation. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

In Tables 9-8, the following notation applies:

- # — Immediate operand
- Dn — Data register operand
- An — Address register operand
- M — Memory operand

Table 9-8. Immediate Instruction Execution Times

Instruction	Size	op #, Dn	op #, An	op #, M
ADDI	Byte, Word	8(2/0)	—	12(2/1)+
	Long	14(3/0)	—	20(3/2)+
ADDQ	Byte, Word	4(1/0)	4(1/0)*	8(1/2)+
	Long	8(1/0)	8(1/1)	12(1/2)+
ANDI	Byte, Word	8(2/0)	—	12(2/1)+
	Long	14(3/0)	—	20(3/1)+
CMPI	Byte, Word	8(2/0)	—	8(2/0)+
	Long	12(3/0)	—	12(3/0)+
EORI	Byte, Word	8(2/0)	—	12(2/1)+
	Long	14(3/0)	—	20(3/2)+
MOVEQ	Long	4(1/0)	—	—
ORI	Byte, Word	8(2/0)	—	12(2/1)+
	Long	14(3/0)	—	20(3/2)+
SUBI	Byte, Word	8(2/0)	—	12(2/1)+
	Long	14(3/0)	—	20(3/2)+
SUBQ	Byte, Word	4(1/0)	4(1/0)*	8(1/1)+
	Long	8(1/0)	8(1/0)	12(1/2)+

+Add effective address calculation time.

*Word only.

9.5 SINGLE OPERAND INSTRUCTION EXECUTION TIMES

Tables 9-9, 9-10, and 9-11 list the timing data for the single operand instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

Table 9-9. Single Operand Instruction Execution Times

Instruction	Size	Register	Memory
NBCD	Byte	6(1/0)	8(1/1)+
NEG	Byte, Word	4(1/0)	8(1/1)+
	Long	6(1/0)	12(1/2)+
NEGX	Byte, Word	4(1/0)	8(1/1)+
	Long	6(1/0)	12(1/2)+
NOT	Byte, Word	4(1/0)	8(1/1)+
	Long	6(1/0)	12(1/2)+
Scc	Byte, False	4(1/0)	8(1/1)+*
	Byte, True	4(1/0)	8(1/1)+*
TAS	Byte	4(1/0)	14(2/1)+*
TST	Byte, Word	4(1/0)	4(1/0)+
	Long	4(1/0)	4(1/0)+

+Add effective address calculation time.

*Use nonfetching effective address calculation time.

Table 9-10. Clear Instruction Execution Times

	Size	Dn	An	(An)	(An)+	-(An)	(d ₁₆ , An)	(d ₈ , An, Xn)*	(xxx).W	(xxx).L
CLR	Byte, Word	4(1/0)	—	8(1/1)	8(1/1)	10(1/1)	12(2/1)	16(2/1)	12(2/1)	16(3/1)
	Long	6(1/0)	—	12(1/2)	12(1/2)	14(1/2)	16(2/2)	20(2/2)	16(2/2)	20(3/2)

*The size of the index register (Xn) does not affect execution time.

Table 9-11. Single Operand Instruction Loop Mode Execution Times

Instruction	Size	Loop Continued			Loop Terminated					
		Valid Count, cc False			Valid Count, cc True			Expired Count		
		(An)	(An)+	-(An)	(An)	(An)+	-(An)	(An)	(An)+	-(An)
CLR	Byte, Word	10(0/1)	10(0/1)	12(0/1)	18(2/1)	18(2/1)	20(2/0)	16(2/1)	16(2/1)	18(2/1)
	Long	14(0/2)	14(0/2)	16(0/2)	22(2/2)	22(2/2)	24(2/2)	20(2/2)	20(2/2)	22(2/2)
NBCD	Byte	18(1/1)	18(1/1)	20(1/1)	24(3/1)	24(3/1)	26(3/1)	22(3/1)	22(3/1)	24(3/1)
NEG	Byte, Word	16(1/1)	16(1/1)	18(2/2)	22(3/1)	22(3/1)	24(3/1)	20(3/1)	20(3/1)	22(3/1)
	Long	24(2/2)	24(2/2)	26(2/2)	30(4/2)	30(4/2)	32(4/2)	28(4/2)	28(4/2)	30(4/2)
NEGX	Byte, Word	16(1/1)	16(1/1)	18(2/2)	22(3/1)	22(3/1)	24(3/1)	20(3/1)	20(3/1)	22(3/1)
	Long	24(2/2)	24(2/2)	26(2/2)	30(4/2)	30(4/2)	32(4/2)	28(4/2)	28(4/2)	30(4/2)
NOT	Byte, Word	16(1/1)	16(1/1)	18(2/2)	22(3/1)	22(3/1)	24(3/1)	20(3/1)	20(3/1)	22(3/1)
	Long	24(2/2)	24(2/2)	26(2/2)	30(4/2)	30(4/2)	32(4/2)	28(4/2)	28(4/2)	30(4/2)
TST	Byte, Word	12(1/0)	12(1/0)	14(1/0)	18(3/0)	18(3/0)	20(3/0)	16(3/0)	16(3/0)	18(3/0)
	Long	18(2/0)	18(2/0)	20(2/0)	24(4/0)	24(4/0)	26(4/0)	20(4/0)	20(4/0)	22(4/0)

9.6 SHIFT/ROTATE INSTRUCTION EXECUTION TIMES

Tables 9-12 and 9-13 list the timing data for the shift and rotate instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

Table 9-12. Shift/Rotate Instruction Execution Times

Instruction	Size	Register	Memory*
ASR, ASL	Byte, Word	$6+2n(1/0)$	$8(1/1)+$
	Long	$8+2n(1/0)$	—
LSR, LSL	Byte, Word	$6+2n(1/0)$	$8(1/1)+$
	Long	$8+2n(1/0)$	—
ROR, ROL	Byte, Word	$6+2n(1/0)$	$8(1/1)+$
	Long	$8+2n(1/0)$	—
ROXR, ROXL	Byte, Word	$6+2n(1/0)$	$8(1/1)+$
	Long	$8+2n(1/0)$	—

+Add effective address calculation time.

n is the shift or rotate count.

* Word only.

Table 9-13. Shift/Rotate Instruction Loop Mode Execution Times

Instruction	Size	Loop Continued			Loop Terminated					
		Valid Count cc False			Valid Count cc True			Expired Count		
		(An)	(An)+	-(An)	(An)	(An)+	-(An)	(An)	(An)+	-(An)
ASR, ASL	Word	18(1/1)	18(1/1)	20(1/1)	24(3/1)	24(3/1)	26(3/1)	22(3/1)	22(3/1)	24(3/1)
LSR, LSL	Word	18(1/1)	18(1/1)	20(1/1)	24(3/1)	24(3/1)	26(3/1)	22(3/1)	22(3/1)	24(3/1)
ROR, ROL	Word	18(1/1)	18(1/1)	20(1/1)	24(3/1)	24(3/1)	26(3/1)	22(3/1)	22(3/1)	24(3/1)
ROXR, ROXL	Word	18(1/1)	18(1/1)	20(1/1)	24(3/1)	24(3/1)	26(3/1)	22(3/1)	22(3/1)	24(3/1)

9.7 BIT MANIPULATION INSTRUCTION EXECUTION TIMES

Table 9-14 lists the timing data for the bit manipulation instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

Table 9-14. Bit Manipulation Instruction Execution Times

Instruction	Size	Dynamic		Static	
		Register	Memory	Register	Memory
BCHG	Byte	—	8(1/1)+	—	12(2/1)+
	Long	8(1/0)*	—	12(2/0)*	—
BCLR	Byte	—	10(1/1)+	—	14(2/1)+
	Long	10(1/0)*	—	14(2/0)*	—
BSET	Byte	—	8(1/1)+	—	12(2/1)+
	Long	8(1/0)*	—	12(2/0)*	—
BTST	Byte	—	4(1/0)+	—	8(2/0)+
	Long	6(1/0)*	—	10(2/0)	—

+Add effective address calculation time.

* Indicates maximum value; data addressing mode only.

9.8 CONDITIONAL INSTRUCTION EXECUTION TIMES

Table 9-15 lists the timing data for the conditional instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format.

Table 9-15. Conditional Instruction Execution Times

Instruction	Displacement	Branch Taken	Branch Not Taken
Bcc	Byte	10(2/0)	6(1/0)
	Word	10(2/0)	10(2/0)
BRA	Byte	10(2/0)	—
	Word	10(2/0)	—
BSR	Byte	18(2/2)	—
	Word	18(2/2)	—
DBcc	cc true	—	10(2/0)
	cc false	10(2/0)	16(3/0)

9.9 JMP, JSR, LEA, PEA, AND MOVEM INSTRUCTION EXECUTION TIMES

Table 9-16 lists the timing data for the jump (JMP), jump to subroutine (JSR), load effective address (LEA), push effective address (PEA), and move multiple registers (MOVEM) instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format.

Table 9-16. JMP, JSR, LEA, PEA, and MOVEM Instruction Execution Times

Instruction	Size	(An)	(An)+	-(An)	(d ₁₆ ,An)	(d ₈ ,An,Xn)+	(xxx) W	(xxx).L	(d ₈ PC)	(d ₁₆ , PC, Xn)*
JMP	—	8(2/0)	—	—	10(2/0)	14(3/0)	10(2/0)	12(3/0)	10(2/0)	14(3/0)
JSR	—	16(2/2)	—	—	18(2/2)	22(2/2)	18(2/2)	20(3/2)	18(2/2)	22(2/2)
LEA	—	4(1/0)	—	—	8(2/0)	12(2/0)	8(2/0)	12(3/0)	8(2/0)	12(2/0)
PEA	—	12(1/2)	—	—	16(2/2)	20(2/2)	16(2/2)	20(3/2)	16(2/2)	20(2/2)
MOVEM M → R	Word	12+4n (3+n/0)	12+4n (3+n/0)	—	16+4n (4+n/0)	18+4n (4+n/0)	16+4n (4+n/0)	20+4n (5+n/0)	16+4n (4+n/0)	18+4n (4+n/0)
	Long	24+8n (3+2n/0)	12+8n (3+2n/0)	—	16+8n (4+2n/0)	18+8n (4+2n/0)	16+8n (4+2n/0)	20+8n (5+2n/0)	16+8n (4+2n/0)	18+8n (4+2n/0)
MOVEM R → M	Word	8+4n (2/n)	—	8+4n (2/n)	12+4n (3/n)	14+4n (3/n)	12+4n (3/n)	16+4n (4/n)	—	—
	Long	8+8n (2/2n)	—	8+8n (2/2n)	12+8n (3/2n)	14+8n (3/2n)	12+8n (3/2n)	16+8n (4/2n)	—	—
MOVES M → R	Byte/ Word	18(3/0)	20(3/0)	20(3/0)	20(4/0)	24(4/0)	20(4/0)	24(5/0)		
	Long	22(4/0)	24(4/0)	24(4/0)	24(5/0)	28(5/0)	24(5/0)	28(6/0)		
MOVES R → M	Byte/ Word	18(2/1)	20(2/1)	20(2/1)	20(3/1)	24(3/1)	20(3/1)	24(4/1)		
	Long	22(2/2)	24(2/2)	24(2/2)	24(3/2)	28(3/2)	24(3/2)	28(4/2)		

n is the number of registers to move.

*The size of the index register (Xn) does not affect the instruction's execution time.

9.10 MULTIPRECISION INSTRUCTION EXECUTION TIMES

Table 9-17 lists the timing data for multiprecision instructions. The numbers of clock periods include the times to fetch both operands, perform the operations, store the results, and read the next instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format.

The following notation applies in Table 9-17:

Dn — Data register operand
M — Memory operand

Table 9-17. Multiprecision Instruction Execution Times

Instruction	Size	op Dn, Dn	Nonlooped	Loop Mode		
				Continued	Terminated	
				Valid Count, cc False	Valid Count, cc True	Expired Count
Instruction	Size	op Dn, Dn	op M, M*			
ADDX	Byte, Word	4(1/0)	18(3/1)	22(2/1)	28(4/1)	26(4/1)
	Long	6(1/0)	30(5/2)	32(4/2)	38(6/2)	36(6/2)
CMPM	Byte, Word	—	12(3/0)	14(2/0)	20(4/0)	18(4/0)
	Long	—	20(5/0)	24(4/0)	30(6/0)	26(6/0)
SUBX	Byte, Word	4(1/)	18(3/1)	22(2/1)	28(4/1)	26(4/1)
	Long	6(1/0)	30(5/2)	32(4/2)	38(6/2)	36(6/2)
ABCD	Byte	6(1/0)	18(3/1)	24(2/1)	30(4/1)	28(4/1)
SBCD	Byte	6(1/0)	18(3/1)	24(2/1)	30(4/1)	28(4/1)

*Source and destination ea are (An)+ for CMPM and -(An) for all others.

9.11 MISCELLANEOUS INSTRUCTION EXECUTION TIMES

Table 9-18 lists the timing data for miscellaneous instructions. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

Table 9-18. Miscellaneous Instruction Execution Times

Instruction	Size	Register	Memory	Register→ Destination**	Source**→ Register
ANDI to CCR	—	16(2/0)	—	—	—
ANDI to SR	—	16(2/0)	—	—	—
CHK	—	8(1/0)+	—	—	—
EORI to CCR	—	16(2/0)	—	—	—
EORI to SR	—	16(2/0)	—	—	—
EXG	—	6(1/0)	—	—	—
EXT	Word	4(1/0)	—	—	—
	Long	4(1/0)	—	—	—
LINK	—	16(2/2)	—	—	—
MOVE from CCR	—	4(1/0)	8(1/1)+*	—	—
MOVE to CCR	—	12(2/0)	12(2/0)+	—	—
MOVE from SR	—	4(1/0)	8(1/1)+*	—	—
MOVE to SR	—	12(2/0)	12(2/0)+	—	—
MOVE from USP	—	6(1/0)	—	—	—
MOVE to USP	—	6(1/0)	—	—	—
MOVEC	—	—	—	10(2/0)	12(2/0)
MOVEP	Word	—	—	16(2/2)	16(4/0)
	Long	—	—	24(2/4)	24(6/0)
NOP	—	4(1/0)	—	—	—
ORI to CCR	—	16(2/0)	—	—	—
ORI to SR	—	16(2/0)	—	—	—
RESET	—	130(1/0)	—	—	—
RTD	—	16(4/0)	—	—	—
RTE	Short	24(6/0)	—	—	—
	Long, Retry Read	112(27/10)	—	—	—
	Long, Retry Write	112(26/1)	—	—	—
	Long, No Retry	110(26/0)	—	—	—
RTR	—	20(5/0)	—	—	—
RTS	—	16(4/0)	—	—	—
STOP	—	4(0/0)	—	—	—
SWAP	—	4(1/0)	—	—	—
TRAPV	—	4(1/0)	—	—	—
UNLK	—	12(3/0)	—	—	—

+Add effective address calculation time.

+Use nonfetching effective address calculation time.

**Source or destination is a memory location for the MOVEP instruction and a control register for the MOVEC instruction.

9.12 EXCEPTION PROCESSING EXECUTION TIMES

Table 9-19 lists the timing data for exception processing. The numbers of clock periods include the times for all stacking, the vector fetch, and the fetch of the first instruction of the handler routine. The total number of clock periods, the number of read cycles, and the number of write cycles are shown in the previously described format. The number of clock periods, the number of read cycles, and the number of write cycles, respectively, must be added to those of the effective address calculation where indicated by a plus sign (+).

Table 9-19. Exception Processing Execution Times

Exception	
Address Error	126(4/26)
Breakpoint Instruction*	45(5/4)
Bus Error	126(4/26)
CHK Instruction**	44(5/4)+
Divide By Zero	42(5/4)+
Illegal Instruction	38(5/4)
Interrupt*	46(5/4)
MOVEC, Illegal Control Register**	46(5/4)
Privilege Violation	38(5/4)
Reset***	40(6/0)
RTE, Illegal Format	50(7/4)
RTE, Illegal Revision	70(12/4)
Trace	38(4/4)
TRAP Instruction	38(4/4)
TRAPV Instruction	38(5/4)

+ Add effective address calculation time.

* The interrupt acknowledge and breakpoint cycles are assumed to take four clock periods.

** Indicates maximum value.

*** Indicates the time from when RESET and HALT are first sampled as negated to when instruction execution starts.

